



New material can reduce lithium in EV batteries up to 70%

The environmental impact of lithium mining has prompted the search for alternatives to lithium-ion batteries. Microsoft and the US Department of Energy's Pacific Northwest National Laboratory have made a breakthrough in battery technology for electric vehicles (EVs). They discovered a material called N2116, which can reduce lithium usage in EV batteries by up to 70%. The team used advanced AI tools to screen 32 million potential materials, developing a battery prototype with the solid-state electrolyte N2116. This innovation, which powered a lightbulb in testing, represents a significant advancement in solid-state battery technology, offering longer driving ranges, faster charging, and increased safety for EVs. Researchers globally are also exploring alternatives, such as sodium-ion and aluminium-based batteries, highlighting a collective effort towards sustainable energy solutions.

New algo automatically selects the fastest route without obstacles for robots

MIT researchers have developed an algorithm that enables robots to navigate complex environments by efficiently constructing roadmaps of uncertain territories. This innovative method balances quality and computational efficiency, allowing robots to identify secure and effi-



MIT researchers developed an algorithm that can automatically select the best shortcuts for a robot to take on its way to a destination that will reduce the overall travel time while limiting the likelihood that the robot will meet an impassable obstacle (Credits: Jose-Luis Olivares, MIT; iStock)

cient routes, minimise travel time, and discover shortcuts. The algorithm outperforms other approaches in simulations, demonstrating its potential in applications ranging from Mars exploration to

search-and-rescue missions. It simplifies the planning process by selecting only viable shortcuts, reducing unnecessary complexity. The algorithm's effectiveness has been validated in over 100 simulated experiments using an aerial campus map. Future enhancements aim to extend its applicability to more complex challenges and address the mismatch between theoretical graphs and real-world environments.

Bifacial solar panels harvest energy from both sides and are 70% cheaper

The research conducted by a team from the University of Surrey, the University of Cambridge, the Chinese Academy of Sciences, Xidian University, and Zhengzhou University introduces a new approach to solar technology. Dr Jing



Bi-facial solar panel

Zhang from Surrey's Advanced Technology Institute highlighted the panels' ability to harvest sunlight from both sides, increasing energy generation and reducing dependency on the angle of incident light.

They have developed a bifacial solar panel using single-walled carbon nanotubes as electrodes on both sides. These nanotubes, with a diameter of 2.2 nanometres, are thinner than a strand of human DNA. The panels have demonstrated good efficiency, with the back panel producing nearly 97% of the power generated by the front panel. The team's panels, which absorb sunlight from both sides, are 70% cheaper to produce than conventional one-sided solar panels, representing a significant step toward making solar technology more affordable and advancing the field of solar energy.

Salt-grain size wireless chips can help monitor internal body health

Brown University engineers have created microchips as small as grains of salt, revolutionising wireless networks. These



The sensors transmit, receive and decode data from thousands of microelectronic chips that are each no larger than a grain of salt (Photos by Nick Dentamaro/Brown University)

chips efficiently transmit, receive, and decode data, ideal for wearables or implants. The silicon sensors send data as electrical spikes, mimicking brain neurons, conserving energy, and bandwidth. They

are scalable, with each sensor being independent, needing no central coordination. These chips have vast potential for biomedical sensors, especially internal health monitoring. They are energy-efficient and powered wirelessly while transmitting. Building on 'neurograins,' the team tested 78 sensors and simulated decoding data from 8000 primate brains. They aim to optimise for lower power and explore broader applications.